

City Rail Link Benefits Realisation Plan



July 2026
Version 10



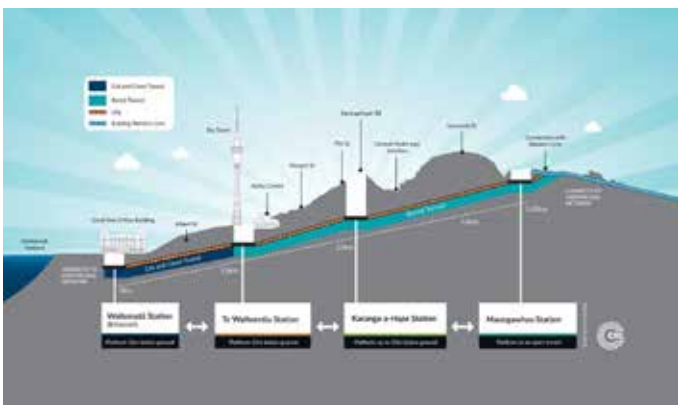
“For the first time, it will be possible to provide a satisfactory public passenger transport service, by a balance between roading, buses and the railway.”

Mayor Sir Dove-Myer Robinson, 1975



“The CRL is the heart of dealing with the growth, with propelling our economy, and creating a future Aucklanders want.”

Mayor Len Brown, 2016



“CRL will deliver Auckland the world-class passenger train network that residents of and visitors to other large cities have taken for granted for generations.”

“...intensification where it makes the most sense—along major transport routes and the CRL.”

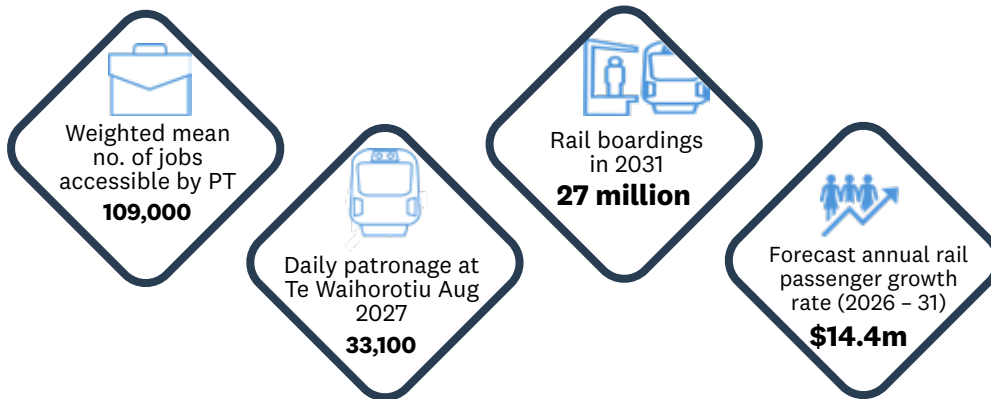
Mayor Wayne Brown, 2026



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Executive Summary



Client and consultant team

City Rail Link (CRL) is New Zealand's largest-ever urban transport project, delivering benefits across Auckland's economy and population.

Auckland Council and Auckland Transport Council engaged the consultancies Invisi, UrbanUmbler and Ascari to understand how CRL reshapes Auckland and to maximise the benefits of this \$5.3bn investment.

When CRL opens...

- 431,000 people will be able to reach Auckland city centre by public transport in 45 mins.
- Rail passenger numbers are forecast to grow by 14.4% annually.
- By 2031, Te Waihorotiu station will be the busiest rail/bus interchange in Auckland.

This benefits realisation plan is to monitor CRL performance, to inform any corrective actions needed and to assess if we are on track to release the benefits identified in the 2015 CRL Business Case.

It contains an updated baseline passenger forecast and 16 measures, informed by existing datasets. This provides the basis for an ongoing CRL monitoring and reporting framework, overseen by Auckland Council's governing body.

This is just the start of a long story. Periodic monitoring will be necessary to measure progress and identify ways to realise the full potential performance and benefits of CRL.

Interventions may include prioritising development near stations and other highly-accessible locations, optimising bus and ferry services and complementary infrastructure investments.

Public transport gives people the freedom of the city. By enabling access to local destinations and the wider region, CRL will enable Auckland – and Aucklanders – to flourish.

Sponsor's Foreword



*Barry Potter, Director Resilience and Infrastructure, Auckland Council
Chair CRL Joint Sponsorship Group*

Auckland's City Rail Link (CRL) has been nearly a century in the making. Its imminent completion reflects central and local government commitment to a more connected, resilient, and prosperous Tāmaki Makaurau.

CRL is more than a transport project; it is a game-changing addition to our largest city; a catalyst for wider urban development. Never before has New Zealand seen metro rail of this quality and scale.

What will CRL mean for the Auckland region? And how can we obtain the best value from our investment?

The CRL Benefits Realisation Plan seeks to answer these questions using data. It sets out a transparent, evidence-based framework to assess how CRL is set to reshape Auckland's transport system and enable further urban development outcomes.

By taking a neutral and rigorous approach to assessing benefits, we can provide the public, businesses, and decision makers with a clear and objective view of CRL.

By monitoring and reporting on its performance, we can maximise the effectiveness of CRL and the wider public transport network it supports.

By using award-winning best practice approaches, developed from British, Irish and Australian examples, we can set a new standard for understanding public transport in New Zealand.

We look forward to working with our partners, stakeholders, and communities as we continue to reshape the future of Auckland's transport system and leverage the opportunities enabled by CRL.

City Rail Link: A city-shaping project



Auckland's City Rail Link consists of twin 3.45 km tunnels below Auckland, with two new city centre stations and enhancements to Waitematā station.

By turning Waitematā (formerly Britomart) into a through-station, CRL will double rail capacity in Auckland and improve public transport access throughout the city.

CRL is more than a set of tunnels and stations. It also includes:

- New trains
- Third main line (Wiri-Westfield)
- Rail Network Rebuild
- New buses
- New bus services
- Streetscape improvements
- Changes to Auckland Unitary Plan.

These inseparable changes will place CRL at the heart of a substantially-improved public transport system.

This report examines what this is likely to mean for Auckland.



How to use the CRL Benefits Realisation Plan

Use it to understand:

- what we have measured
- how to use the data
- what this means for Auckland.

The City Rail Link Benefits realisation Plan contains:

- Fully-updated CRL passenger forecasts
- Baseline predicted travel and use patterns once CRL has opened
- Access maps illustrating multiple measures
- Explanations of what this data means
- Advice for next steps and long-term monitoring.

The measures focus on whether the intended outcomes of CRL are being achieved in practice.

They are underpinned by high-quality technical analysis and informed by international best practice.

Details are contained in the technical notes and appendices:

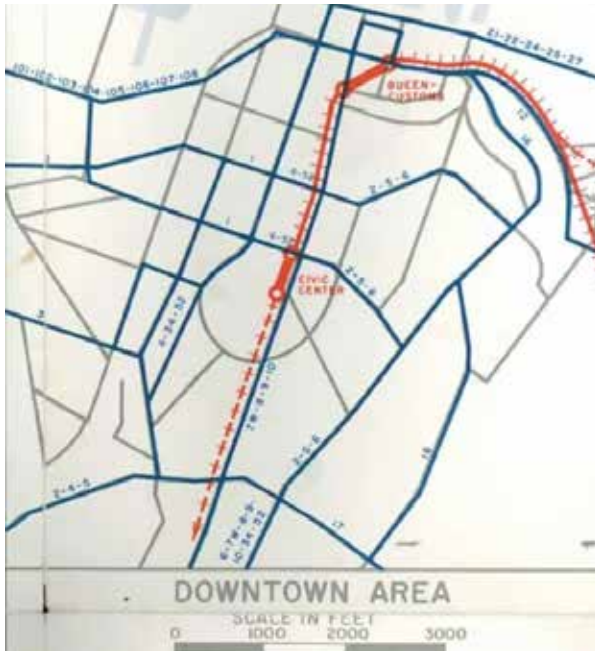
- Review of 2015 CRL Business Case
- Technical notes on:
 - CRL MSM forecasts
 - Access mapping
 - Generating orthogonal experimental design for MSM model runs
 - OLS + log-transformed regression
 - Walk parameters for station catchment modelling
 - Monte Carlo methodology.

Glossary of terms

Term	Meaning
AGS	Auckland Growth Strategy 1.1
GTFS	General Transit Feed Specification – detailed public transport operational information used to inform analysis
H3	Hexagonal grid mapping system. See p. 11
Monte Carlo	Probabilistic forecasting method. See page 21
MSM	Macro Strategic Model (used by Auckland Forecasting Centre)
OSM	Open Street Map – for street and path network. See P. 11
R	Software package used to analyse CRL data
R5R	Rapid Realistic Routing on Real-world & Reimagined networks
Rail Catchment	Places within 15-minute walk of railway station. This measurement incorporates gradients and street network connectivity

City Rail Link: A brief history

Auckland's first passenger rail terminus opened in 1885. Since the 1920s, Auckland has sought to build railway tunnels and stations beneath its city centre streets.



De Leuw Cather Auckland rail proposal, 1963

Previous CRL plans

1923: Morningside Deviation: Rail tunnel from Morningside to Beach Road, with station at Auckland Town Hall. Rejected in 1930.

1949: Morningside Deviation (refined): Electrified loop on similar route to 1923 plans. Cancelled in 1954.

1963: De Leuw Cather proposal: Rail tunnel with 2x new stations at Customs St and Town Hall, to open by 1970. This was not taken forward.

1972: Auckland Rapid Rail Transit Plan ("Robbie's Rapid Rail"): City centre tunnel, plus new lines to Airport and North Shore; funding refused in 1975.



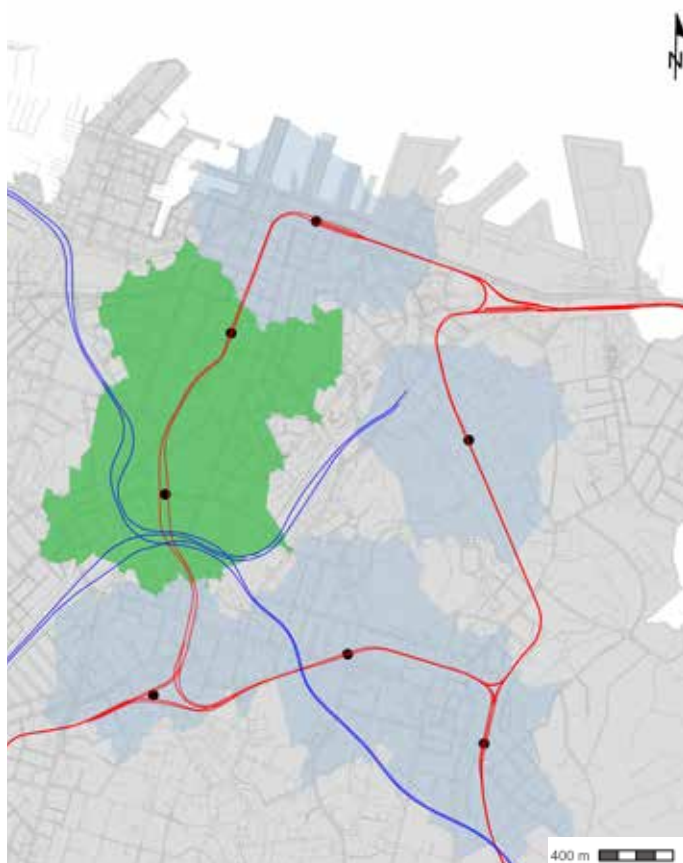
New electric trains at Britomart, 2015

Auckland's Rail Renaissance

Passenger rail in Auckland struggled through the 1990s. The opening of Britomart station in 2003 returned trains to the heart of Auckland. The Britomart precinct quickly developed. Passenger numbers grew rapidly.

Auckland Transport introduced new electric trains, rebuilding stations such as Ōtāhuhu and New Lynn. Bus and train services and ticketing were integrated, delivering further growth.

Britomart station was a terminus and was limited to 20 trains per hour. While designed for conversion into a through-station, a rail route via central Auckland was not yet designated.



Expanded *walking catchments* due to CRL

CRL route designated

In 2008, Westfield obtained resource consent to develop the 1970s Downtown Shopping Centre with retail and office blocks. A large underground car park would have blocked any future rail tunnels, rendering CRL unviable.*

Fully aware of the importance of CRL, Auckland Regional Transport Authority (ARTA) challenged Westfield's consent.

The Minister of Finance found "it was in the long-term public interest to secure and protect the CBD tunnel route". A deal was thus reached to build CRL.

*Further information is available in the 2015 CRL Business Case; available online.

Timeline of City Rail Link

2009	ARTA and KiwiRail begin planning CRL
2010	CRL business case released by Auckland Council, led by Mayor Len Brown
2011	CRL designation notified, safeguarding route. Land acquisition begins
2013	Auckland Council directs Auckland Transport to seek designation for CRL
2014	Enabling works design contracts tendered for Britomart, Downtown and Albert St
2015	Auckland Council approves internal CRL business case and \$400m Enabling Works
2016	Auckland Council and Crown agree \$3.4bn CRL budget (50:50 split) Physical construction begins
2017	Responsibility for delivery transferred to newly-established City Rail Link Ltd (CRL)
2019	Main works contracts successfully tendered; Link Alliance established to deliver tunnels and stations; rail systems added in 2020 Budgeted costs for CRL increase to \$4.41bn
2023	Budget revision to \$5.5bn approved
2025	Civil construction complete; first test train runs
2026	Testing and commissioning. City Rail Link opens in second half of 2026

Building begins

This deal opened the door to CRL planning, design, funding and enabling works. The CRL Business Case (2015) confirmed 50% Crown funding, enabling CRL construction to start in 2016.



Ceremony to mark the start of CRL construction, 2016

CRL Business Case (2015)

The 2015 CRL Business case set out a comprehensive set of benefits, drawn directly and indirectly from travel time savings.

- **Direct benefits** include \$1.3bn travel time savings for public transport users.
- **Indirect benefits** include \$899m of agglomeration benefits resulting from increased access to Auckland city centre.

The 2026 CRL Benefits Realisation Plan (BRP) updates the benefit-cost ratio of CRL and seeks to understand how CRL transforms access in Auckland.

This report develops work previously done for the Ministry of Transport. It is also a response to requests from the Treasury and the Office of the Auditor General (OAG) for a deeper understanding of CRL benefits.

The OAG has requested:

- Clear ownership
- Measurable outcomes
- Coordinated reporting.

This report directly addresses these requests. It seeks to inform long-term monitoring and maximise the city-wide benefits of CRL.



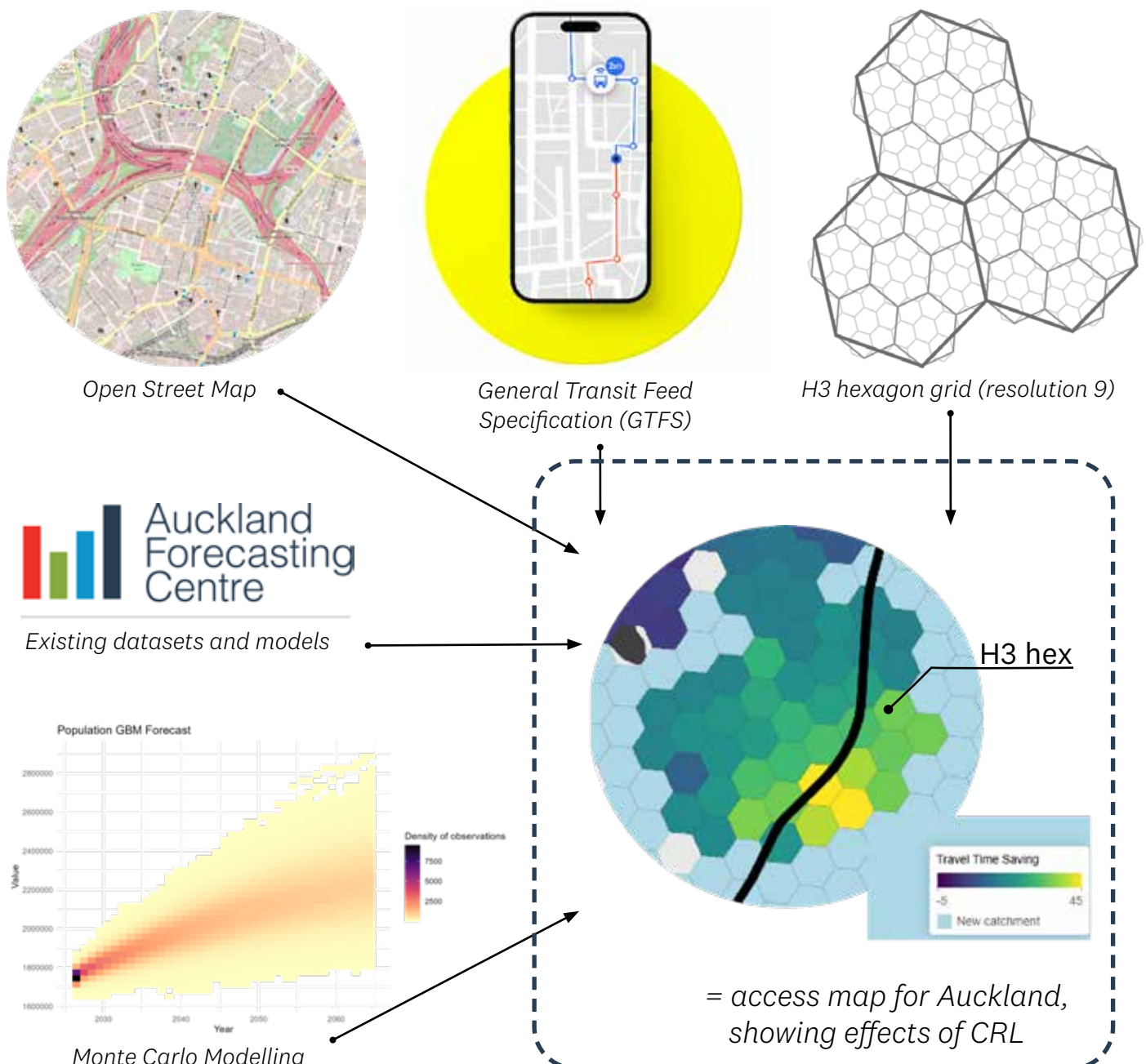
Infographic from 2015 CRL Business Case summarising benefits of City Rail Link

Methodology

The City Rail Link BRP uses a city-wide access model built in r5r, using data from OpenStreetMap, General Transit Feed Specification, existing Auckland Transport models, census data and land use data, mapped to a H3 hexagon grid.

For this project, we have mapped all modes of public transport in Auckland, plus walking. We have not yet mapped cycling, nor park-and-ride trips to stops/stations.

Further information is provided in the technical reports.



Next steps - actions & monitoring

The CRL BRP provides high-quality forecasts of travel time savings, rail demand determinants and city-wide access.

Auckland can use this information to make the most of the benefits available from CRL.

Demand forecasting

We have used Monte Carlo forecasting to identified seven variables that determine rail passenger growth rates.

Variable		% elasticity
1	Population	+1.53
2	Work-from-home rates	-0.125
3	Time-of-use charges	+0.0013
4	Train frequency	+0.206
5	Rail fares	-0.26
6	Time spent on train	-0.70
7	Car parking charges	+0.11

“% elasticity” = how much a 1% change in this variable will affect rail patronage growth:

- + means an increase in this variable will **increase** rail patronage
- means an increase in this variable will **decrease** rail patronage

To increase rail patronage, we can:

1. **Increase** the population within reach of railway stations
2. **Increase** train frequencies
3. **Decrease** journey times & fares

Access Mapping

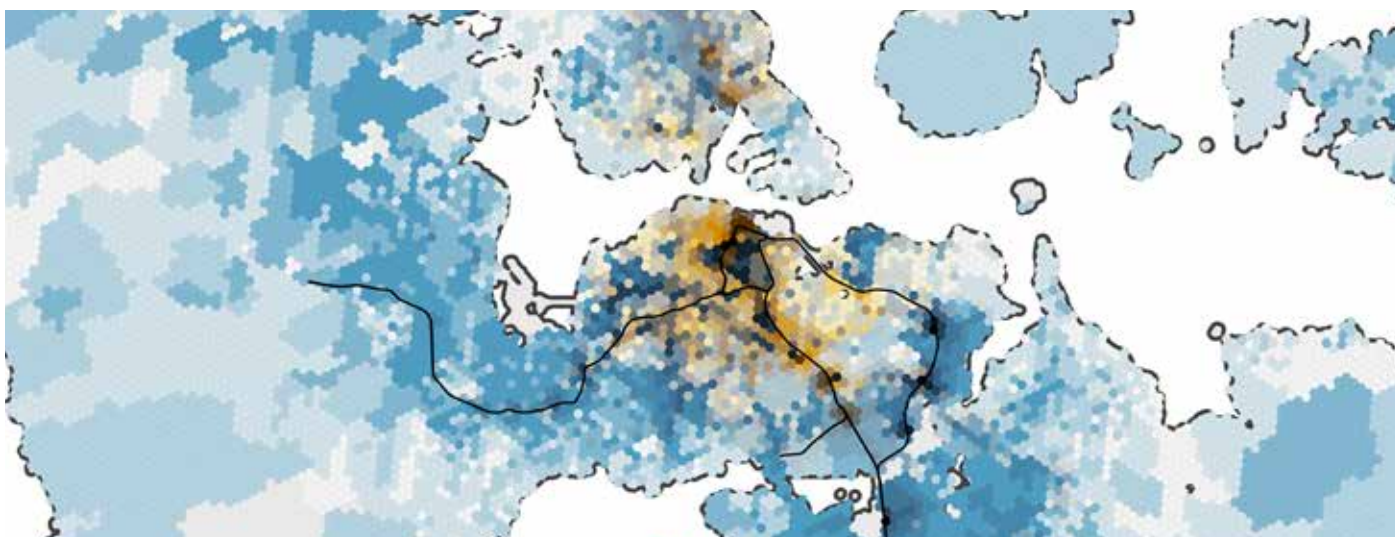
We know how many people can access each hex area in Auckland by public transport and walking, and that higher public transport patronage improves services, via the Mohring effect.

To maximise access, we should:

1. Prioritise development in highly-accessible locations, through zoning and other measures
2. Engage with landowners to unlock high-value sites in low-value uses
3. Maximise pedestrian network connectivity near stations (first-and-final leg)
4. Develop public transport networks to address connectivity gaps



Everywhere-to-everywhere access map (p. 37)



Deprivation and access to jobs by public transport in Auckland (see p. 55)

Knowledge sharing

This report is informed by a robust technical base. Material includes:

- Research findings from applying the Auckland-wide access model for public transport and walking
- Interactive H3 access maps
- Monte Carlo modelling forecasts and graphical user interface (GUI)

To share knowledge, we should:

1. Produce custom journey planning maps (for councillors, executives, etc) to show bespoke scenarios
2. Host interactive maps and monitoring dashboard online
3. Integrate the findings from the CRL BRP reports into Auckland Council planning processes
4. Develop the Auckland-wide access model to include:
 - cycling and micromobility
 - park-and-ride
5. Produce a complete, city-wide view of transport and land use integration in Auckland.

Long-term monitoring

CRL will re-shape Auckland over the long term. As such, it is important to measure these impacts over the long term to enable and inform better decisions. Some measures (e.g. changes in deprivation) will take years to appear.

We can use the CRL BRP analysis, baseline and findings to:

1. Monitor measures in CRL BRP (p.16)
2. Augment CRL BRP with other measures and datasets
3. Integrate analysis to understand long-term effects of CRL on re-shaping Auckland.

Therefore, we should:

1. Use data to inform long-term strategic analysis
2. Ensure that a dedicated, long-term resource is made available to support CRL monitoring
3. Use this information to implement measures to maximise CRL benefits
4. Fulfil long-term monitoring requirements of Treasury and Office of the Auditor General (OAG).

Step 1:

Updating the baseline: 2015-26

There are two steps to the 2026 City Rail Link Benefits Realisation Plan.

The first step was to update the passenger baseline predictions to reflect changes since the business case was written in 2015.

Changes since 2015:

- Public transport improvements, including New Bus Network and Climate Action Transport Targeted Rate (CATTR) routes and services
- COVID-19 pandemic, reducing weekday travel demand and increasing remote/hybrid working
- Planning policy and regulatory changes, including Auckland Unitary Plan (2016) and Plan Change 120 (forthcoming) supporting higher development densities, particularly near rapid transit stations
- Legislation to enable time-of-use charging
- Development of Macro Strategic Model (MSM); new Auckland-wide transport model; highly-detailed, internationally peer-reviewed.

Compared to 2015 CRL Business Case:

- Opening day passenger forecasts are ~30% lower
- Passenger growth rates are broadly similar (14.4%)
- The CRL BRP will measure passenger growth relative to these forecasts.
- Full calculations are contained in the accompanying CRL BRP Technical Report.

CRL BRP baseline settings

- **Annualisation factor** = 213
- **Work-from-Home rate** = 18%
- **Land use** = Auckland Growth Scenario (AGS) 23 v1.1
- **Train Plan** = Auckland Signalling Capacity Improvements (ASCI) Aug 2025
- **Time of use charging model** = MoT 2020 The Congestion Question; 2031- City Centre cordon toll; 2041: Strategic Corridors.

Rail patronage: observed & forecast

The **blue** line shows Auckland's annual rail patronage since 1990, with:

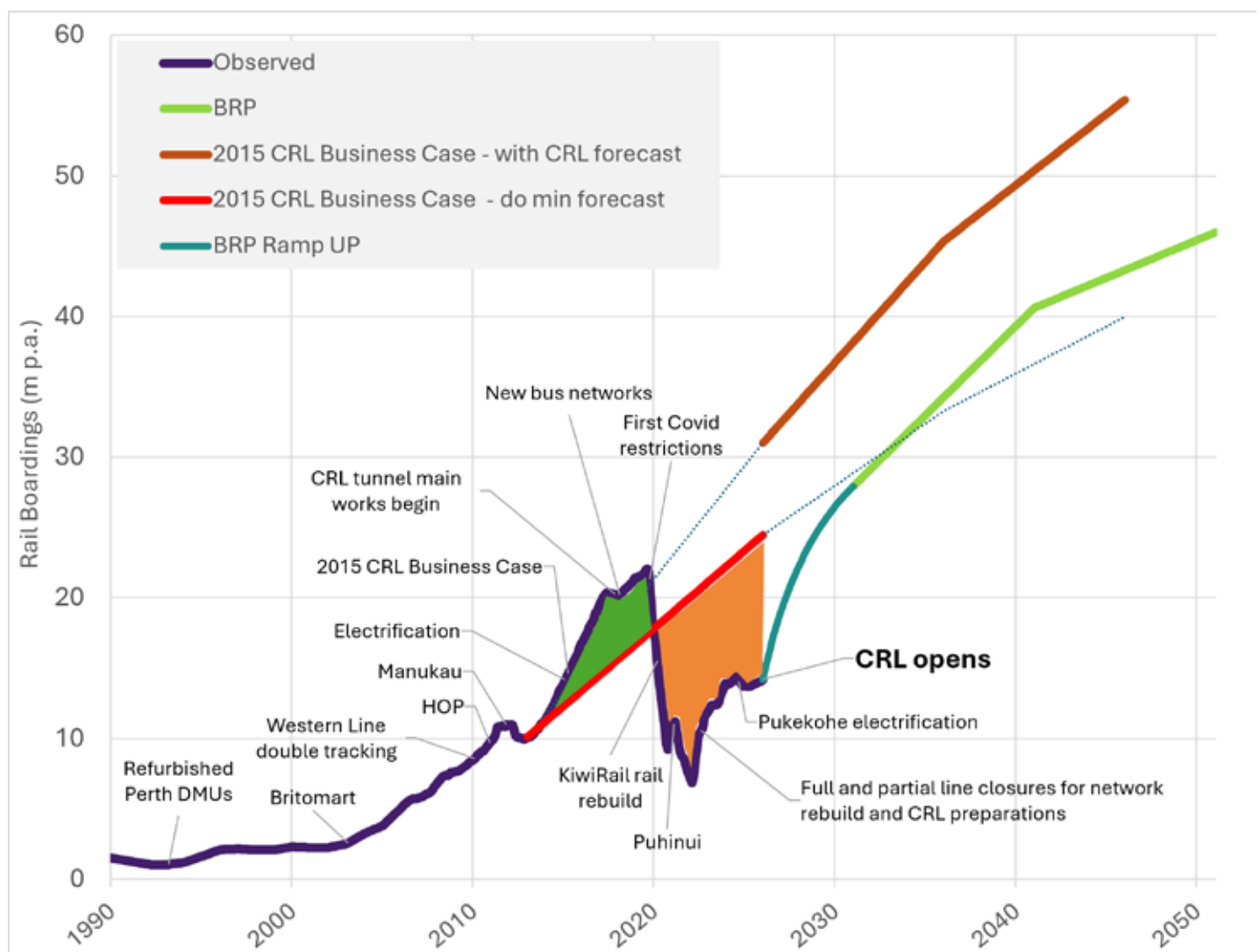
- strong growth after Britomart opened (2003)
- rapid growth with electrification (2013-15)
- Steep decline during Covid-19 pandemic
- Weak recovery due to rail network rebuild and changing travel patterns

The 2015 CRL Business Case forecast:

- shows a do-minimum (i.e. CRL not built) passenger forecast in **red**
- the passenger forecast with CRL is shown in **orange**

The 2026 CRL BRP updates this forecast:

- the first five years are shown in **turquoise**
- forecasts from 2031 onwards are shown in **green**



Step 2:

Expanding the scope

The second step is to expand the scope with additional measures to understand how CRL will reshape Auckland.

There are three categories of benefits:

1. Transport network performance
2. Accessibility
3. Development and density

Each benefit category has four or more measures, all based on publicly-available, high-quality datasets.

These are shown in the table below.

Benefit category	Measures
1. Transport network performance	Total passenger numbers
	Travel times to Midtown
	Change in travel times to Midtown
	City centre stations: Daily patronage
	City centre passenger transfers
	City centre footfall
2. Accessibility	Access to jobs
	Multi-activity access score
	All-Auckland access levels
	Public transport effective job density
	Network benefits of CRL
	% of jobs in rail catchments
3. Development and Density	Population density
	Building consents
	Change in anywhere-to-anywhere PT access
	Deprivation and access to jobs

Expanded Walking Catchment with CRL

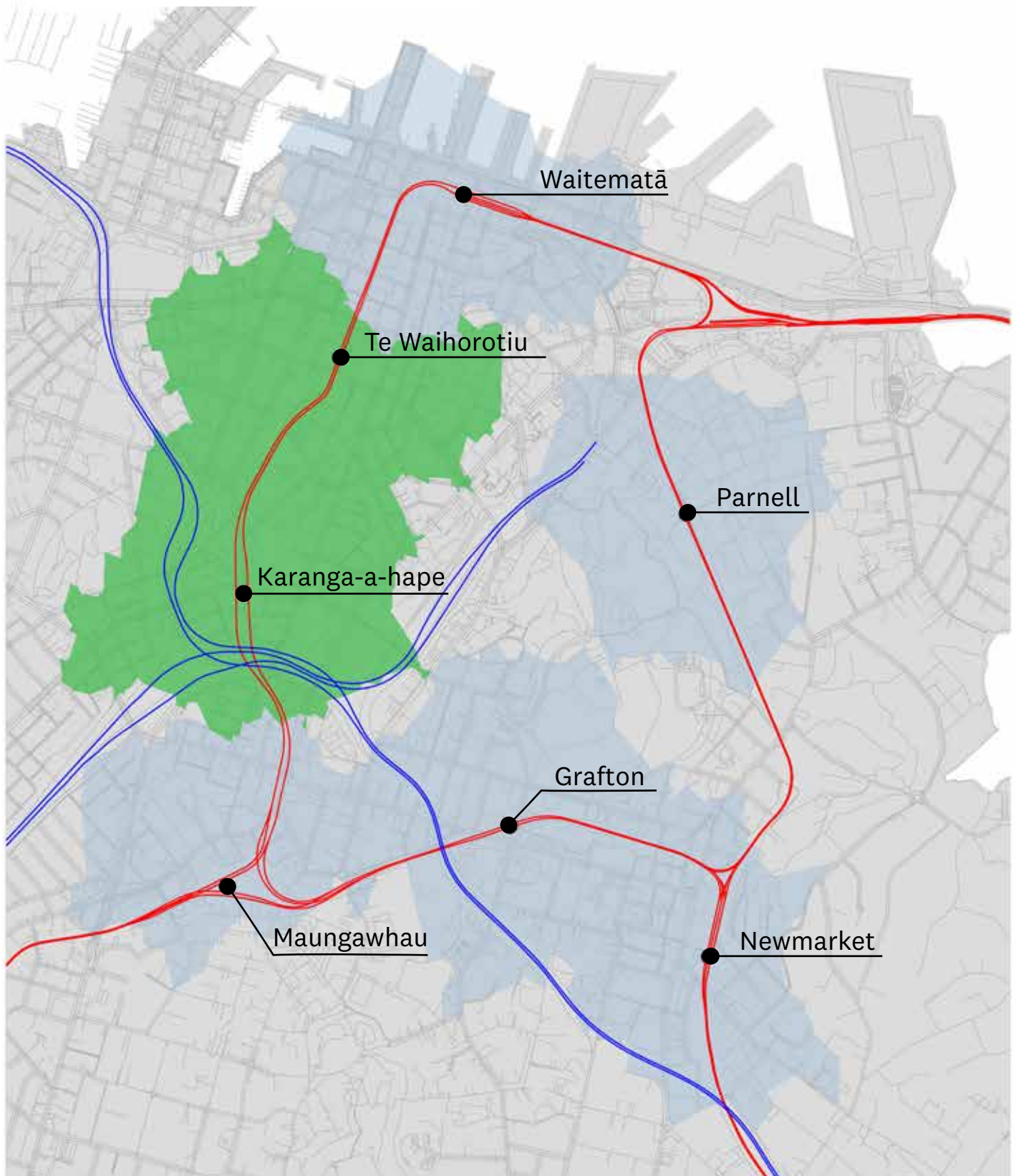
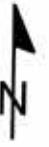
Green area shows additional walkable coverage from new CRL stations

21,000 more city centre residents

17,200 more students

37,000 additional (existing) jobs

} will be within a 12-15 minute walk of the new CRL stations.



400 m

An architectural rendering of the Te Waihorotiu station entrance. The image shows a modern building with a complex, blue and grey facade. A large white rectangular box is overlaid on the upper left portion of the image, containing the text '1. Transport Network Performance' in a bold, dark blue font. Below this, the station entrance is visible, featuring a sign that reads 'Te Waihorotiu'. The entrance area includes a set of stairs, a platform with a yellow and black striped canopy, and a sign indicating '1-2' and 'Platform'. Several people are depicted in the scene, including a woman sitting on a concrete bench in the foreground, a woman pushing a stroller, and various other pedestrians. The overall scene is brightly lit, suggesting a sunny day.

1. Transport Network Performance

The background image shows a modern urban environment. On the left, a tall building with a white, textured facade rises. In the foreground, a wide, paved pedestrian crossing is populated with several people walking. To the right, a dark blue bus with the number 'GB815' and the 'Metro' logo is visible. A traffic light with a green arrow is also present. The overall scene is bright and clear, suggesting a sunny day.

CRL will transform transport in Auckland

Despite its short length, City Rail Link's effects on Auckland's public transport network are far-reaching. CRL increases speed, frequency, reliability and proximity to public transport.

This section measures what CRL will mean for Auckland's public transport network. It covers passenger forecasts, travel time savings, station patronage, city centre transfers and footfall.

What should Auckland do?

Maximise the attractiveness and usability of Auckland's integrated public transport network. This can happen via operational decisions, land-use planning and the design and management of streets and public transport networks.

1. Transport Network Performance

Total Passenger Numbers

What are we measuring?

Growth in patronage due to CRL*.

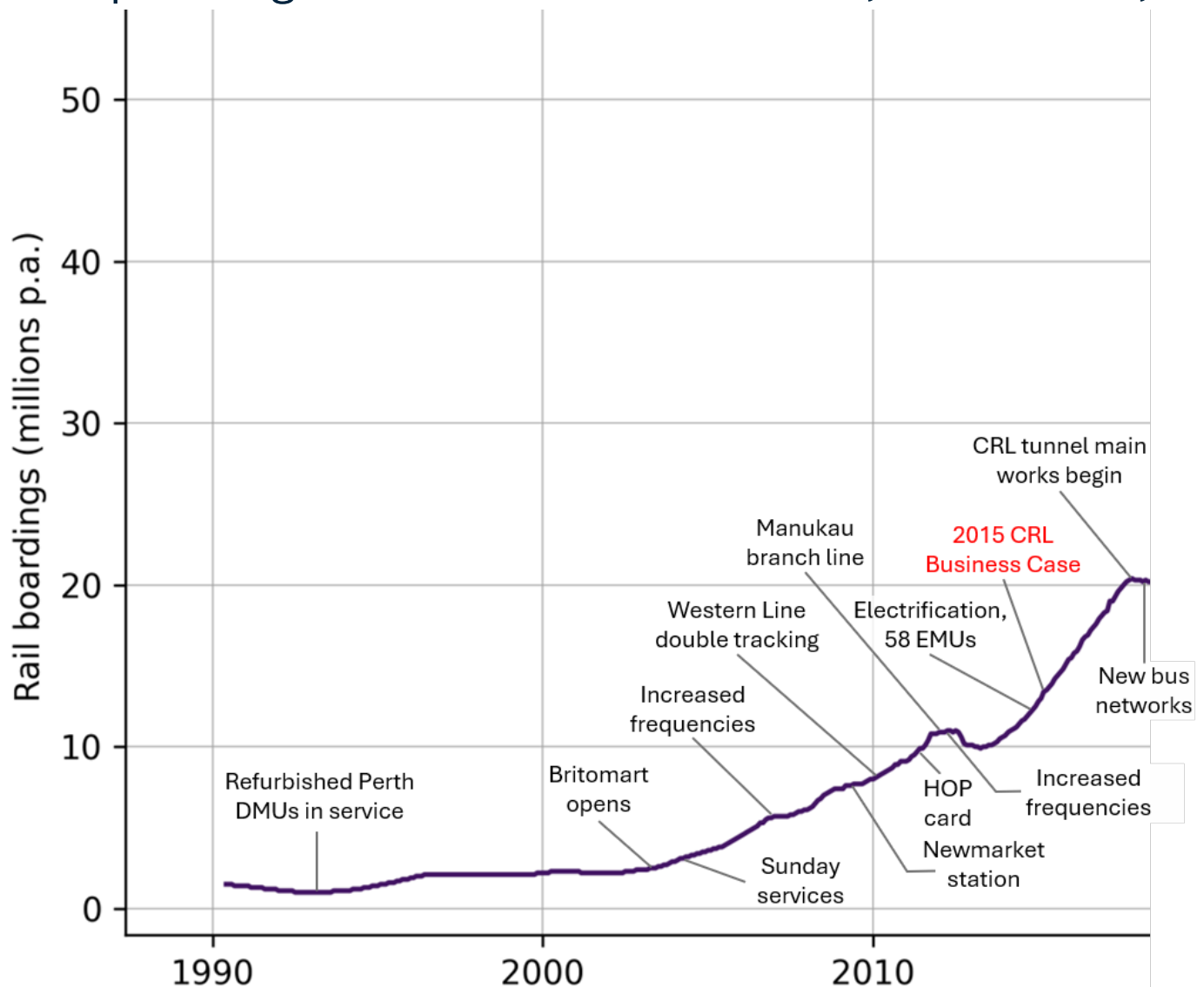
The total number of train journeys in Auckland in a 12-month period, starting from the first month of CRL operations, measured through boarding and alighting at each station.

What is it important?

This will tell us if the growth in passenger numbers reflects forecasts and whether the network reflects passenger requirements.

*an updated graph will include a no-CRL curve, for comparison

Rail patronage and forecasts in Auckland, 1990 – 2050,



What passenger numbers should we expect?

With CRL Forecasts for annual Auckland Metro rail boardings are:

- At CRL opening: 14.2 million
- After 1 year: 19 million
- After 5 years: 27.9 million

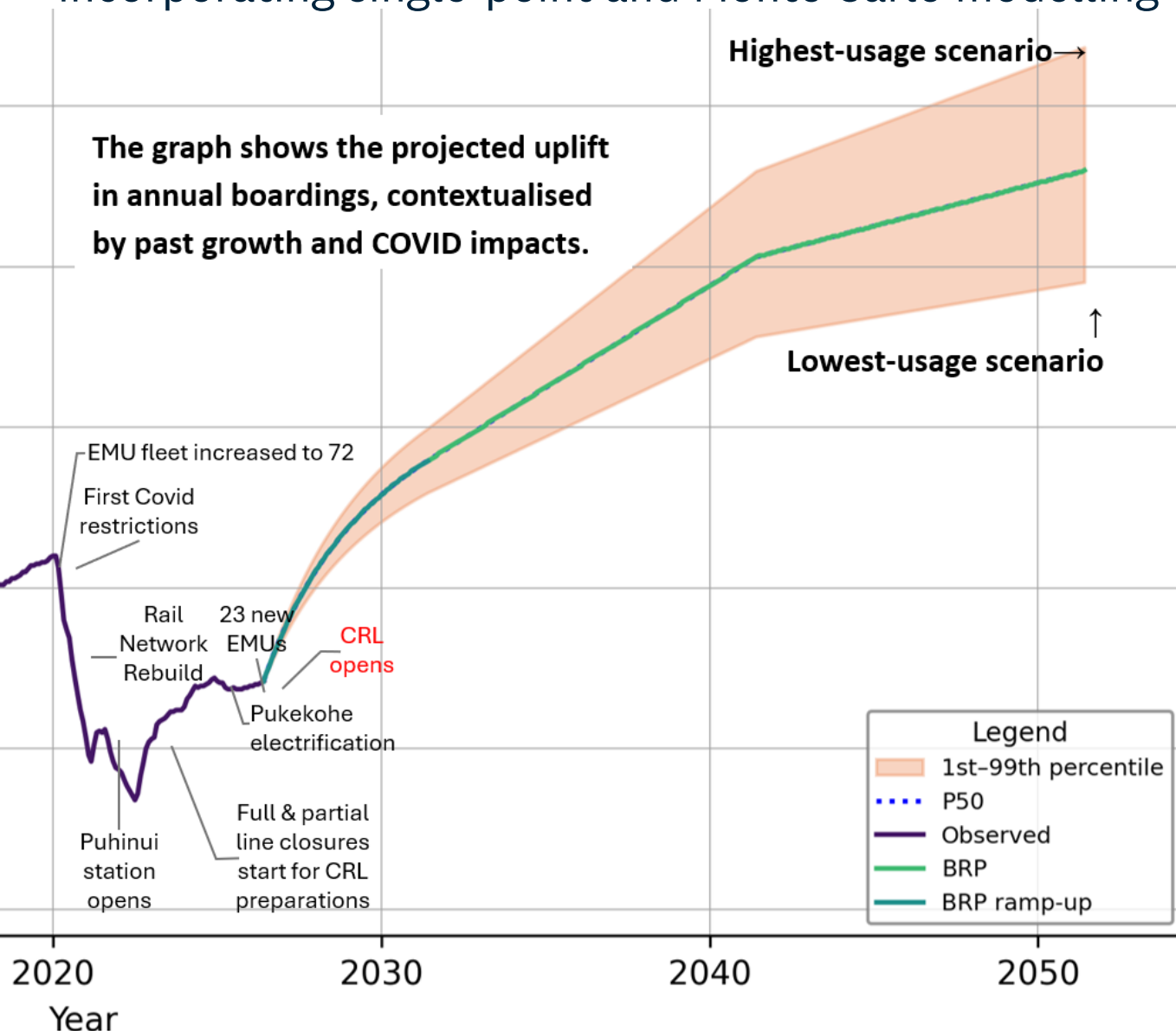
Under the worst-case scenario, rail patronage will double in ten years.

Achieving 27.9 million boardings by 2031 requires a compound average annual growth rate (CAAGR) of 14.4%.

This is comparable to the 14.9% annual growth achieved between December 2012 and December 2017 during earlier network expansion phases.

incorporating single-point and Monte Carlo modelling

Source: ARC, ARTA, AT, UrbanUmbler



1. Transport Network Performance

Public transport travel times to Midtown

79 mins

Median PT travel time
to Midtown in 2025



69 mins

Median PT travel time
to Midtown in 2025

NB: These figures are *unweighted**

What are we measuring?

The total journey time by public transport or walking from origins and destinations across the Auckland urban area to Te Waihorotiu station.

These times **include walking to/ from stations, waiting, and transfers between services.**

A multi-modal trip could go:

walk→**wait**→**bus**→**wait**→**rail**→**walk**

Why is it important?

Travel time improvements underpin the CRL Business Case. If travel times do not improve as much as expected, public transport will be less attractive than forecast. Fewer people will use it and the benefits of CRL will be reduced.

What should we expect?

Public transport travel times to reduce by an average of 10 minutes (unweighted)*.

We should see travel times decrease across areas served by rail and, to a lesser degree, those areas where users can transfer to rail from bus/ferry.

What does the map show?

The map shows travel times by public transport to Te Waihorotiu station in Auckland city centre in the morning peak. It uses information from the Day 1 CRL timetable.

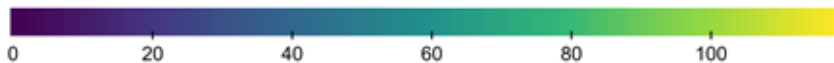
Travel times for key locations are summarised in Appendix [2: [Travel Time for Key Locations](#)]

*unweighted estimates do not account for higher proportions of rail passengers nearer to Te Waihorotiu station. A weighted median PT travel time would be lower than 69 minutes.

Travel times to Te Waihorotiu (CRL Day 1)

Northern Busway
stations provide quick
connections to Midtown.

Travel Time



1. Transport Network Performance

Change in travel times to Midtown

41 mins

Travel time reduction from
Drury to Te Waihorotiu
when CRL opens



22 mins

Travel time reduction
from Kelston to Te Wai-
horotiu when CRL opens

What are we measuring?

The *change in time* taken to reach
● Te Waihorotiu station in central
Auckland once CRL has opened.

Why is it important?

CRL brings travel time savings,
particularly to the west, south and
east of Auckland. These benefits are
widespread because of the way that rail
and bus services are integrated.

What should we expect?

By reducing travel times across much of
Auckland's integrated public transport
system, CRL is likely to increase rail, bus
and ferry patronage.

When sites becomes more accessible
to/from Auckland city centre, we can
expect increases in land values and
development. This will depend on zoning
and other factors.

What does the map show?

- **West Auckland** sees significant travel time reduction. West of New Lynn, travel times fall by around 20 minutes.
- **Whenuapai** receives a new bus service when CRL opens
- **South and East Auckland** see travel time reductions of around 10 minutes. New bus routes (37 and 39) bring extra time savings to Wiri, Puhinui and Highbrook.
- **Drury and Karaka** receive <2hr public transport connections to midtown for the first time.
- **Isthmus and North Shore** travel times are mostly unchanged where buses remain the best public transport mode to Midtown.

Travel times to Te Waihorotiu: CRL Day 1 vs February 2026

Whenuapai
• New bus services

Widespread travel
time reductions
(15-20 minutes)
throughout west
Auckland

Wiri and Puhinui
• New bus routes
(37, 39)

Drury
• New rail station
• New bus services

NB: This analysis does not include
park and ride. This if added, would
increase the catchment of stations
on Auckland's urban fringe

Travel Time Saving



New catchment



1. Transport Network Performance

City centre stations: daily patronage

63,000

Daily city centre CRL station patronage after 12 months



90,000

Daily city centre CRL station patronage forecast in 2031

What are we measuring?

The number of people boarding and alighting at each CRL station in the city centre each day. This includes:

- People who start or end their journey in the city centre
- People who transfer to/from other modes of public transport (i.e. buses and ferries) to reach their final destination.

What should we expect?

Daily use of the four city centre CRL stations will grow as new travel behaviours become established.

Policy changes like time-of-use charging and higher car parking costs will increase CRL patronage further.

Service changes, including longer operating hours, higher speeds and higher frequencies, are also likely to increase patronage.

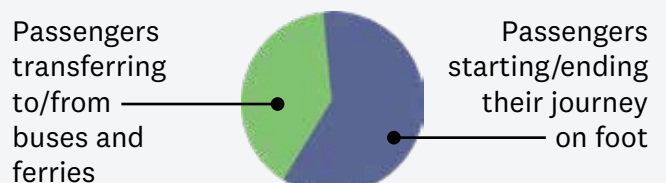
Why is it important?

This will show

- how successfully CRL attracts people into the City Centre
- how well CRL integrates with wider public transport networks, extending CRL benefits beyond immediate rail catchments.

What does the map show?

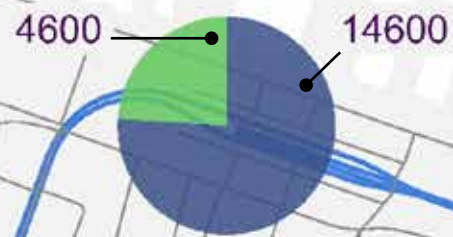
- Daily patronage at each CRL station one year after opening
- Proportion of passengers transferring to buses/ferries vs. walking.



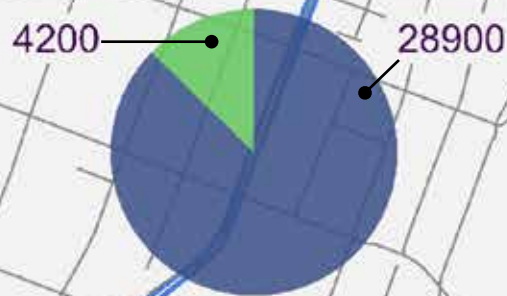
**City centre CRL patronage
12 months after opening**



Waitemata



Te Waihorotiu



Karanga-a-Hape



Maungawhau



Access

- walkup
- transfer

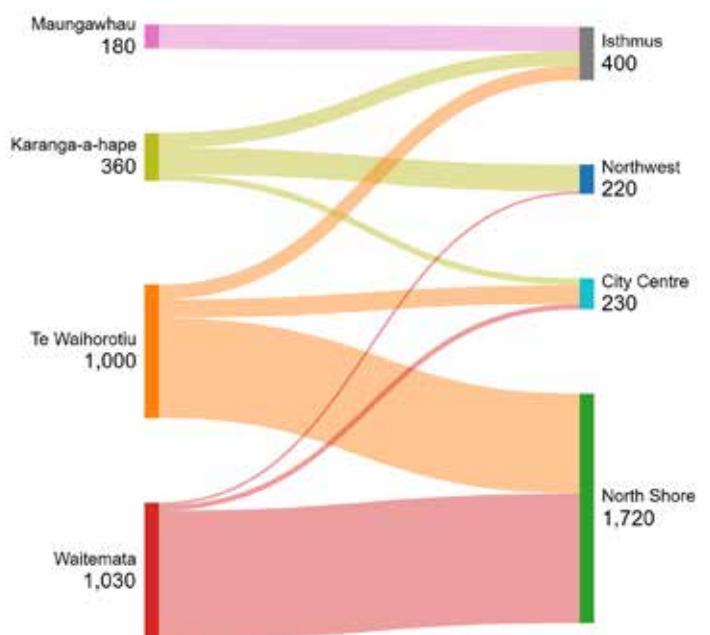
City centre transfers

These are **Sankey diagrams**. They show forecasts for rail passengers transferring to/from buses and ferries at Auckland's four central rail stations in 2026 and 2031.

About 20% of city centre rail passengers will be transferring to/from buses and ferries. More information is provided on page 30.

CRL Transfer Catchments 2026

In 2026, the North Shore provides the highest number of passengers, with Waitematā being the most important station for transfer.

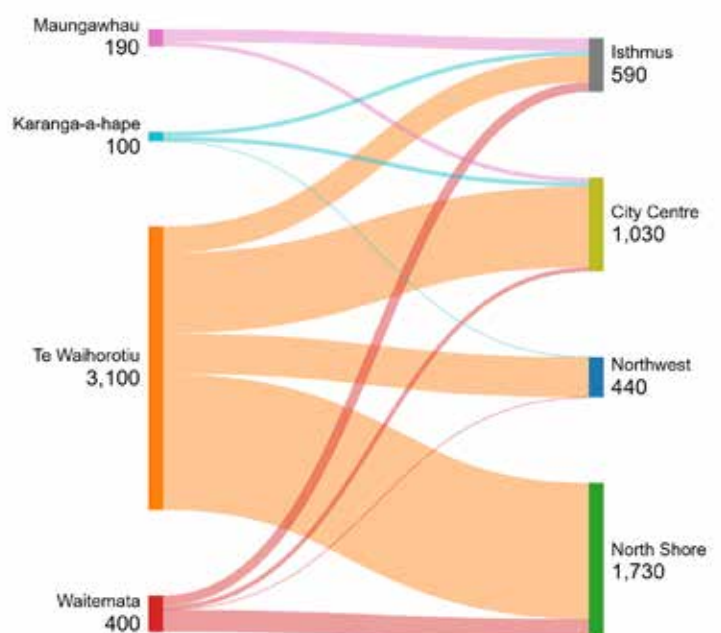


Made at SankeyMATIC.com

CRL Transfer Catchments 2031

By 2031, Te Waihorotiu station becomes the most important transfer station in central Auckland.

Karanga-a-hape is less important for bus transfers due to changes in WX1 bus services.

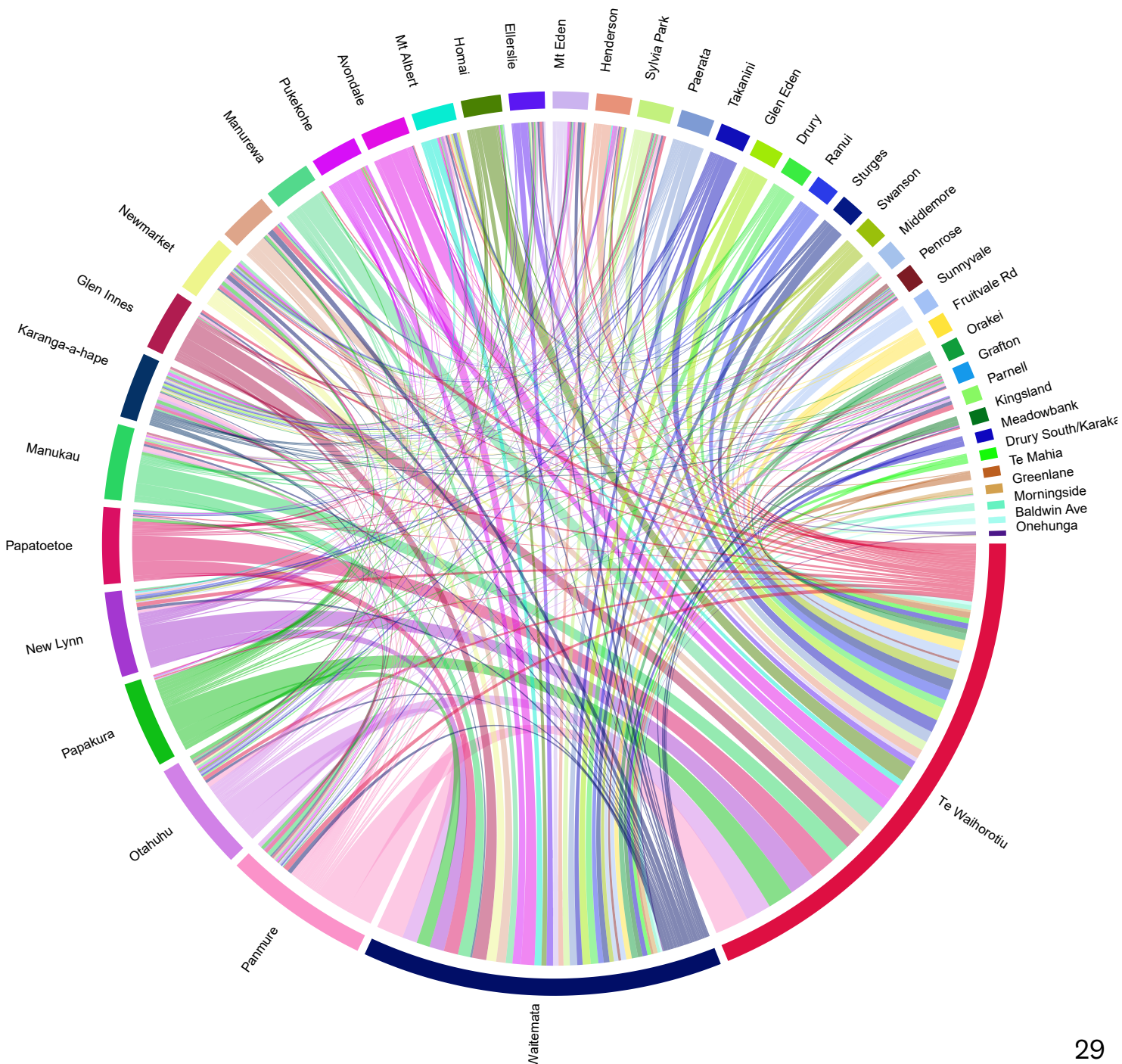


Made at SankeyMATIC.com

Origins and Destinations

This is a chord diagram. It shows forecast boardings and destinations for every railway station in Auckland when CRL is open.

- Passenger numbers are highest for the two main city centre stations, reflecting the highest concentration of people, jobs and destinations
- Next highest are the bus/rail interchanges to the south and west. This shows the importance of Auckland's integrated public transport system
- The top eight stations account for over 50% of passengers
- Chord patterns will change over time as per the adjacent Sankey diagrams.



1. Transport Network Performance

Citywide public transport access

22%

of city centre rail passengers will transfer to other services

...of which...



45%

are travelling to/from the North Shore

What are we measuring?

The onward destinations of CRL passengers in Auckland city centre.

In 2031 20,600 people are forecast to alight at four CRL stations (Karanga-a-hape, Waitematā, Te Waihorotiu and Maungawhau) in the morning peak.

Of those:

- **16,000 (78%)** will walk to their final city centre destination
- **700 (3%)** transfer to another train
- **3,800 (19%)** alight and transfer to other modes – buses and ferries.

Why is it important?

The \$5.3 billion invested in CRL brings benefits well beyond the walkable catchments of railway stations.

70% of transfer passengers will head to destinations far outside the rail network catchment. CRL enables faster, more integrated journeys across Auckland's wider urban area.

What should we expect?

Catchments will expand as new rapid transit connections come online, including:

- Northwest Busway
- Eastern Busway
- Airport-to-Botany corridor

This will enable faster, more connected journeys in Auckland

What does the map show?

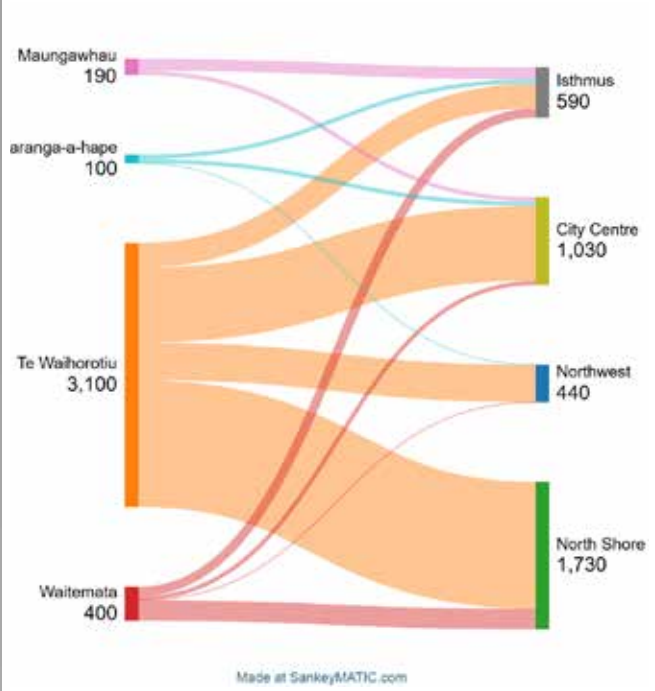
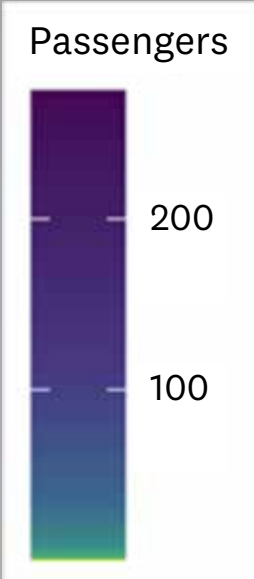
By integrating rail, bus and ferry services, CRL will enable multi-modal trips throughout Auckland:

Short onward trips include Wynyard Quarter, the University precinct, and the Hospital district

Longer journeys include:

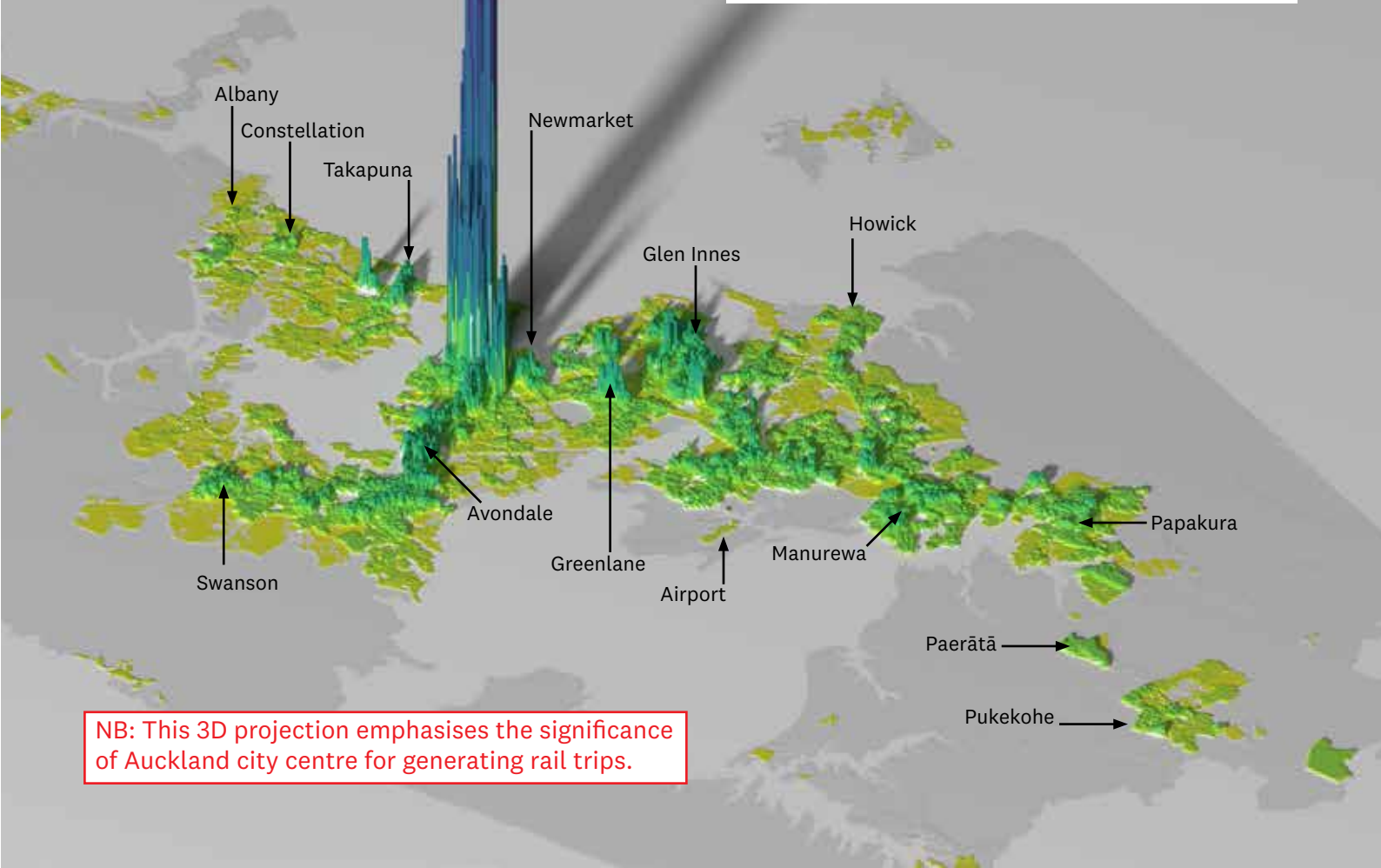
- Northwest
- Southeast (Botany, Howick, Pakuranga)
- Southwest (Māngere, Favona, Manurewa)
- North Shore (Silverdale, Albany, Takapuna, Akoranga, Birkenhead).

Diagram to show origins and destinations of CRL trips.



3,800 passengers forecast to transfer in am peak in 2031.

Destinations	%
Northwest	11%
Isthmus	16%
City Centre	27%
North Shore	45%
Other	1%



NB: This 3D projection emphasises the significance of Auckland city centre for generating rail trips.

2. Accessibility



Accessibility is crucial to Auckland's economic well-being.

38% of New Zealand's economic activity occurs in Auckland. Its GDP per capita is around 10% higher than the national average, reflecting agglomeration economies and high job density.

This figure is much lower than for comparable cities. Sydney has a 59% GDP per capita premium over regional New South Wales, while London's GDP per capita is 71% higher than the UK average. Public transport enables people and businesses, to connect and specialise, boosting productivity.

This section measures what CRL will mean for access in Auckland. It covers locations of jobs, city centre footfall and overall city-wide accessibility.

What should Auckland do?

Use CRL to maximise access, agglomeration and productivity. Auckland's economic growth and well-being will depend on its public transport system connecting people with destinations.

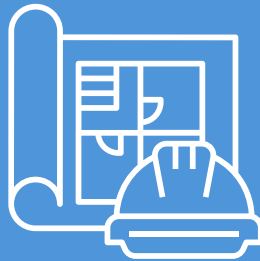
Accessibility will underpin prosperity.

2. Accessibility

Public transport access from anywhere to anywhere

431,000

People can access Waitematā station by public transport in 45 mins



428,000

People can access Otāhuhu station by public transport in 45 minutes

What are we measuring?

The average (mean) number of people that can access each destination zone via walking and public transport in the morning peak within 45 minutes.

Why is it important?

The more people can reach a given location, the more relevant it is for the city. Highly accessible locations can support a greater variety of activities and are prime candidates for enabling density done well.

What should we expect?

Highly accessible locations are likely to attract development and investment interest. The actual level of development will depend on planning policy and economic factors.

What does the map show?

The most accessible points in Auckland are generally located at railway stations and on rapid and frequent bus routes connecting to stations.

- **Waitematā station** is the single most accessible hex zone (431,000 people in 45 mins). This reflects strong public transport connections and a large population within walking distance.
- **Ōtāhuhu** is the second most accessible hex (428,000 people in 45 mins). Rail and bus connections are concentrated a narrow isthmus at Ōtāhuhu, making it easily reachable from south Auckland and the central Isthmus.

People accessible within 45 minutes (CRL Day 1)

Waitematā station

Otāhuhu station

NB: This map shows the levels of accessibility with a 45 minute journey time. Shorter/longer journey times would produce different results.

People accessible



2. Accessibility

Access to jobs by public transport

84,000

The mean number of jobs accessible in 2026



109,000

The weighted number of jobs accessible in 2026

What are we measuring?

The number of jobs accessible by public transport and walking within 45 minutes.

We divide Auckland into thousands of hexagonal zones; an **H3 grid**.

Each zone is multiplied by the percentage of Auckland's population that lives there. This provides a **weighted mean**, reflecting Auckland's population distribution.

When CRL opens, the average number of jobs accessible in 2026 from zones within the Auckland Urban Area is forecast to be **84,000**.

If weighted by population, it is forecast to be **108,868**.

Why is it important?

Access to jobs helps determine where people want to live, the range of employment available, productivity and land values.

What should we expect?

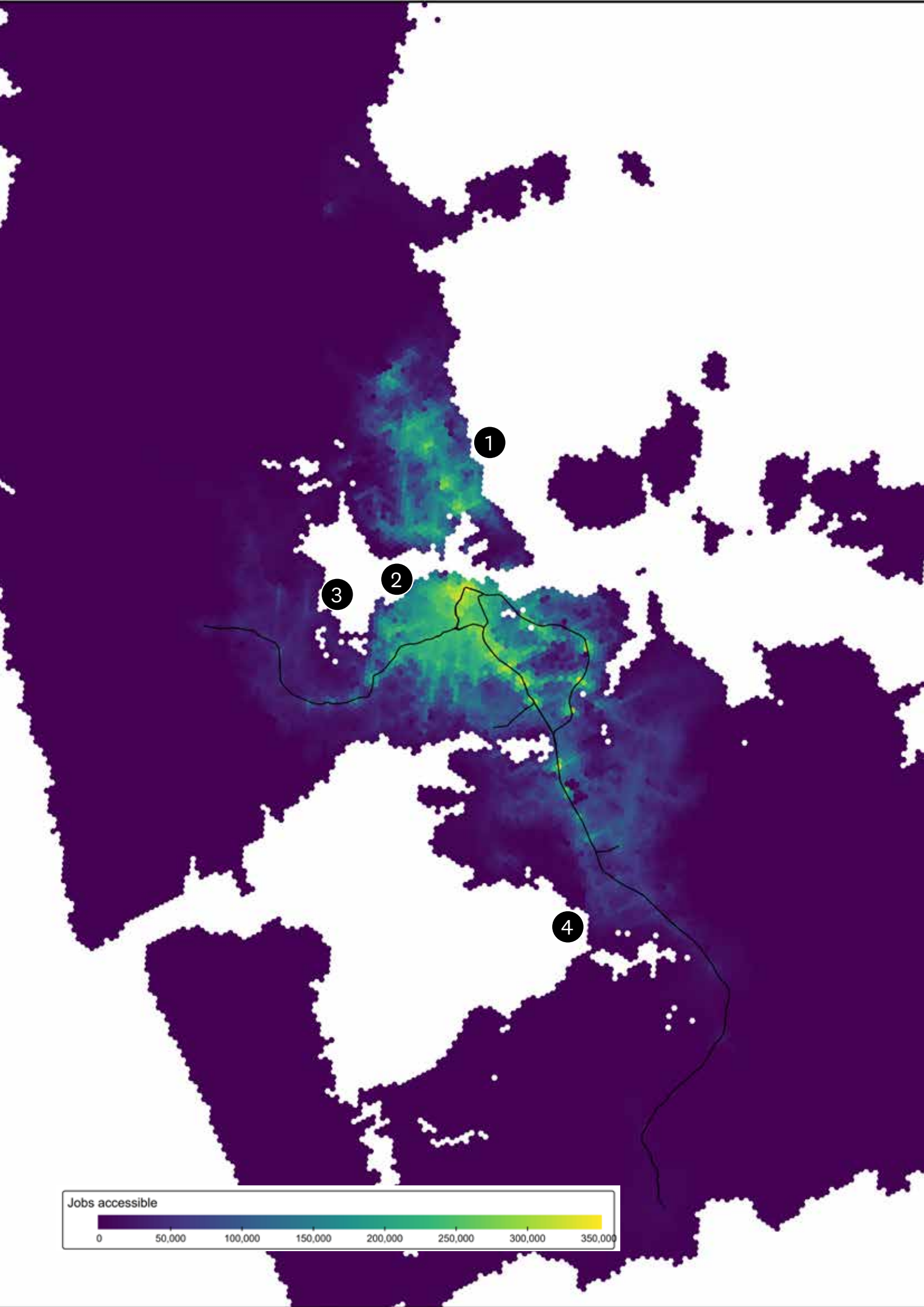
At a population level, people prefer quicker journeys to work and are willing to pay for that benefit.

Land values increase when people are able to access more jobs. This stimulates development.

What does the map show?

The number of jobs accessible from each zone within the Auckland Urban Area.

- 1 North Shore:** Fast, frequent bus services in Onewa Rd, Glenfield Rd and Northern Busway connect people to jobs
- 2 Isthmus:** Consistently high access to jobs, especially along southern rail corridor
- 3 West:** Inner west has best job access; limited beyond Avondale
- 4 South:** Job access is highest at/near railway stations



2. Accessibility

Multi-activity access score



What are we measuring?

The multi-activity **access score** is the total time required to access (by public transport and walking):

50,000 jobs	a doctor's surgery
a grocery store	a university
a school	3x restaurants
a hospital	green space

A lower access score is better because this means a person can access many destinations in a given period of time.

Why is it important?

The Access Score shows how well Auckland's public transport connects people to everyday destinations.

Access scores indicate the levels of local service provision and PT access. They show how convenient it is to live in a particular neighbourhood without needing daily access to a car.

What should we expect?

CRL reduces mean travel times, making more places accessible within a given travel time budget.

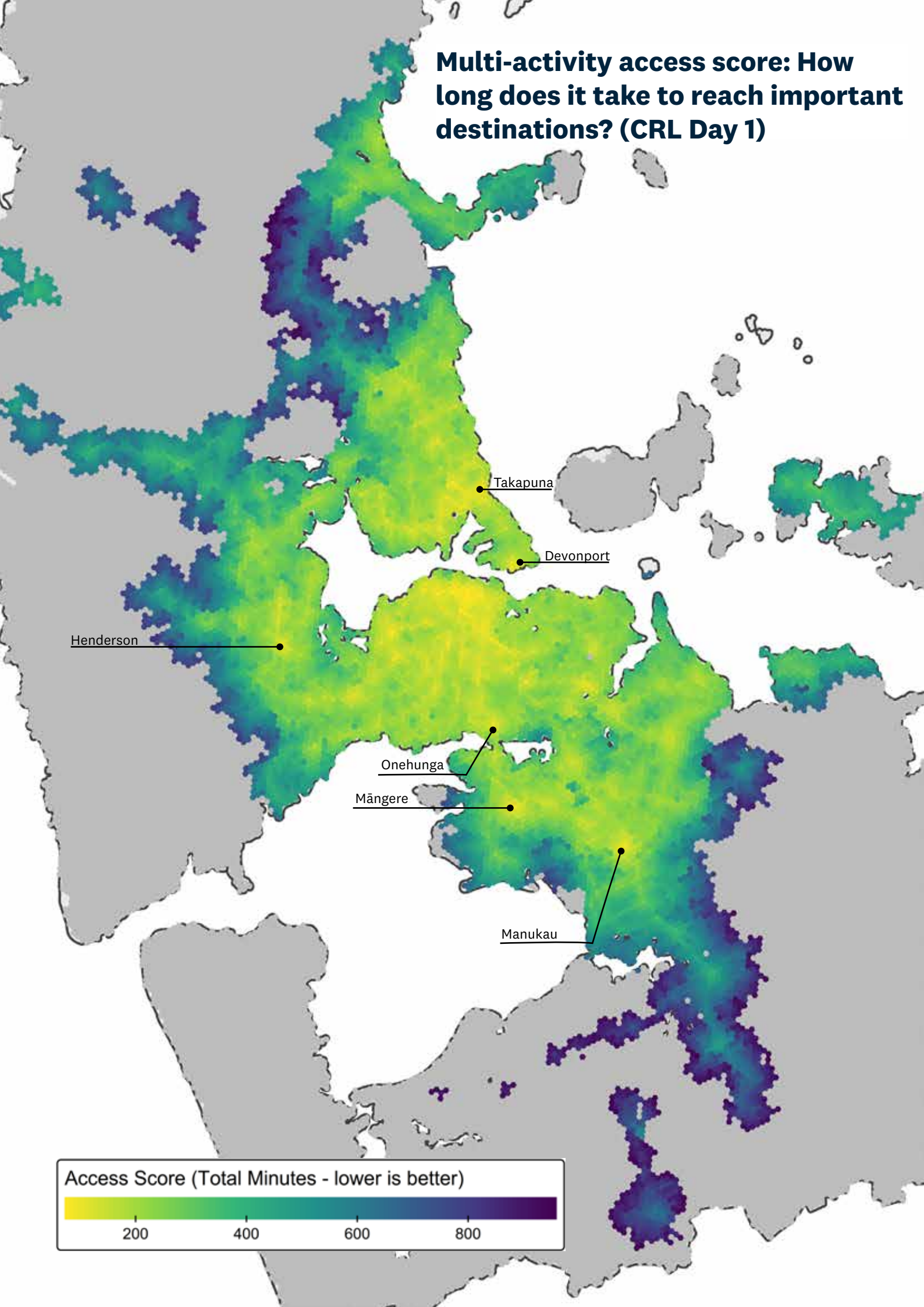
If public transport increases the catchment size of an existing centre, development may occur/increase

What does the map show?

The map shows the access score across the Auckland Urban Area.

- We see a wider, more polycentric geographic distribution of high access scores than we saw with access to jobs (p. 31)
- Local, regional and metropolitan centres score well throughout Auckland, reflecting the distribution of social infrastructure
- Centres with relatively poor access to the rest of Auckland (e.g. Māngere, Devonport) can still have good local access scores.

Multi-activity access score: How long does it take to reach important destinations? (CRL Day 1)



Access Score (Total Minutes - lower is better)



2. Accessibility

Proportion of jobs in rail catchments



What are we measuring?

The change in the proportion of overall regional jobs located in areas served by rail.

Why is it important?

The change in proportion of jobs in areas served by CRL will indicate whether businesses benefit from improvements to the rail network.

Cities with employment centred on dense passenger rail networks tend to be more productive.

This is because rail brings many people and businesses close together from across a wide catchment, connecting people, jobs, specialisations and ideas.

This process – agglomeration – is a key source of productivity and economic well-being in cities.

What should we expect?

Based on the Auckland Growth Scenario (AGS), we expect there to be approximately 726,600 jobs across the Auckland region on CRL Day 1, of which % are in rail catchments.

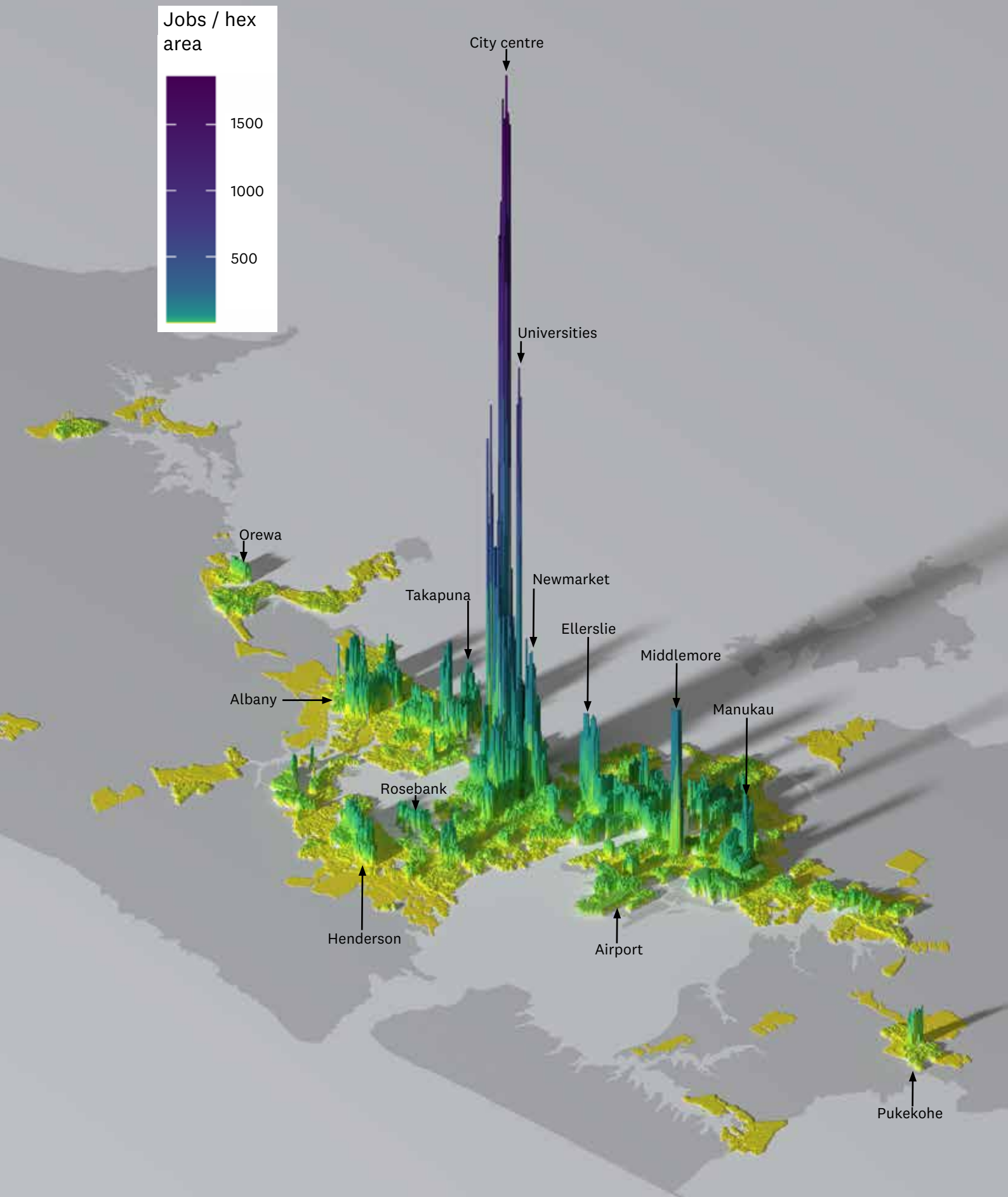
Job growth is forecast to be highest at **existing nodes**. Significant/above-trend levels of job growth may indicate new areas of agglomeration.

What does the map show?

The map shows the distribution of jobs in Auckland in 2026. A 3D plot is used to emphasise how employment is concentrated at public transport stations.

Jobs are concentrated in areas with public transport access; city centre, Newmarket, Manukau, Henderson and Takapuna.

Distribution of jobs in Auckland (CRL Day 1)



2. Accessibility

Public transport effective job density (PT EJD)

Public Transport Effective Job Density (EJD) tells us how many jobs we can afford to access. It combines:

- the *number of jobs* in an area with
- the *travel cost* needed to access them
- weighted by how many people travel that way.*

What are we measuring?

Public Transport Effective Job density (PT EJD) combines

- number of jobs per census zone
- the generalised (time and money) costs of accessing them by public transport*
- A weighting, determined by how many people travel in this way

What should we expect?

Quicker travel times and higher population and employment will increase EJD in areas with public transport.

What does the map show?

The map illustrates the PT EJD by zone across Auckland. This analysis is based on Macro-Strategic Model (MSM) outputs.

Why is it important?

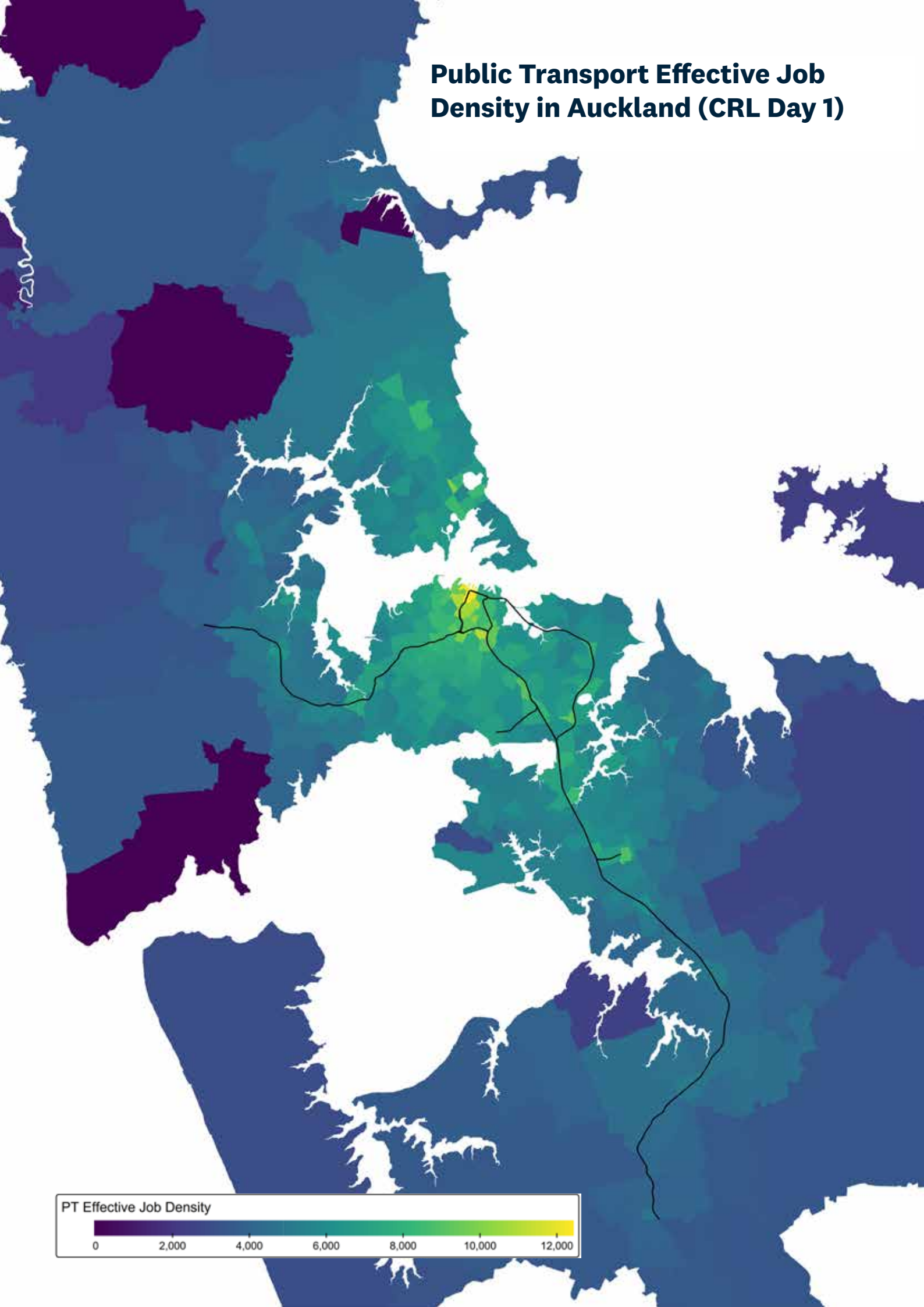
Higher PT EJD is associated with greater prosperity because jobs tend to be concentrated in areas of cities with highest public transport connectivity.

This relationship reflects agglomeration economies. When public transport brings more people and businesses closer together, people connect and learn from one another. This is where ideas and innovation happen, and how businesses specialise.

PT EJD is highest where there are high concentrations of public transport connections and jobs.

- PT EJD is highest by far in Auckland city centre
- Newmarket, the city fringe and, on the North Shore, Smales Farm have high PT EJD
- Industrial locations e.g. Highbrook and East Tamaki have *high levels of employment, but low PT EJD*.

**Public Transport Effective Job
Density in Auckland (CRL Day 1)**



PT Effective Job Density



1. Transport Network Performance

Increases in city centre footfall



What are we measuring?

The number of people walking to and through key areas of Auckland's city centre, measured by automatic counters placed in the city centre.

We will monitor the median hourly pedestrian count over all counting sites on an ongoing basis.

All pedestrian counters are located close to future CRL stations.

Why is it important?

Pedestrian counting helps us to understand whether the opening of the CRL leads to more - or fewer - potential customers for city centre businesses.

It also shows how changes in footfall are distributed across different locations.

What should we expect?

We expect to see an overall increase in pedestrian activity once the CRL opens and the city centre becomes more accessible to all Aucklanders.

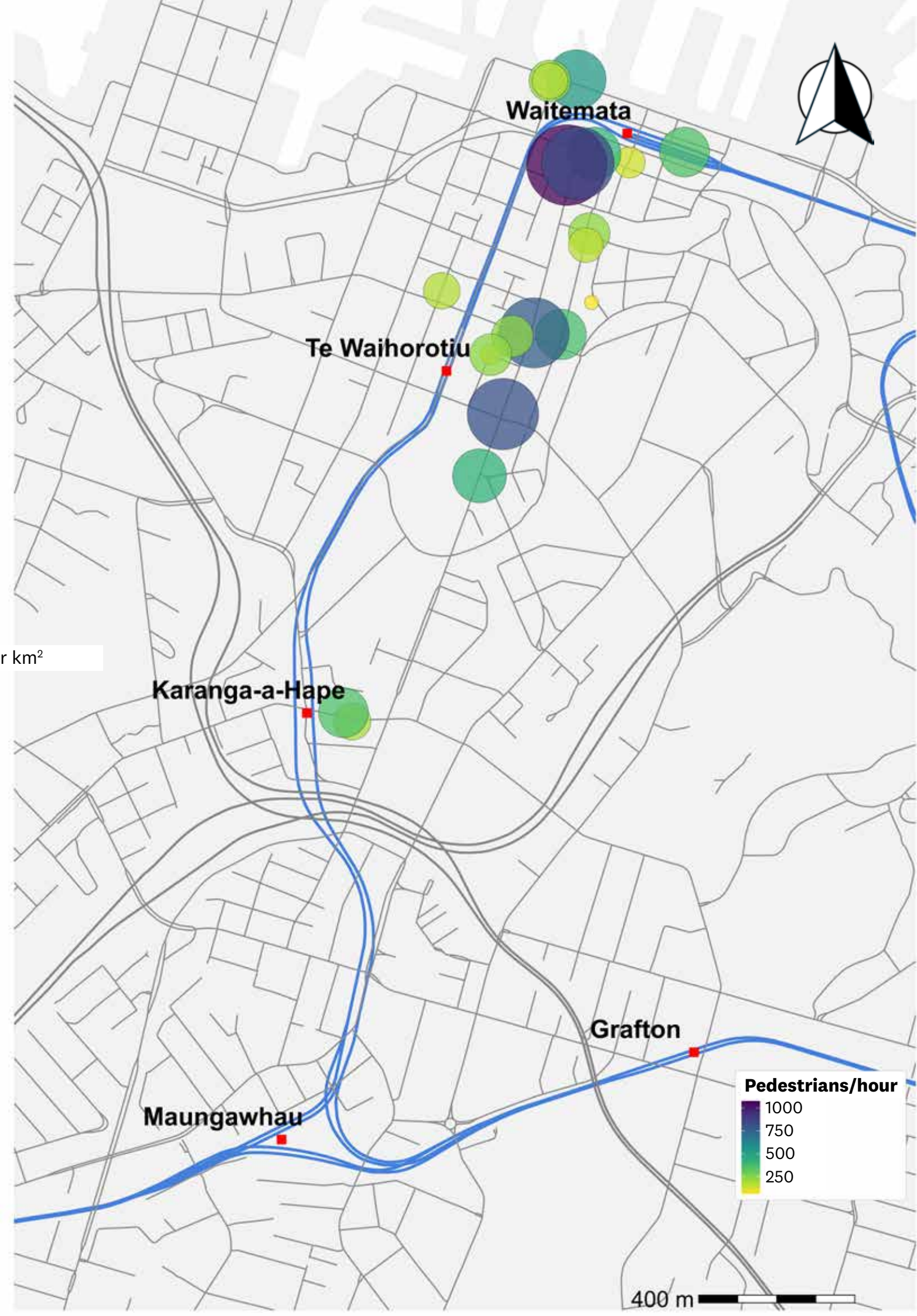
What does the map show?

The map shows the current median hourly pedestrian flows at each location.

Pedestrian counters

Pedestrian counters are managed by city centre business association Heart of the City.

More information is available on their website: <https://city-centre/results-and-statistics/pedestrian-counts>



3. Development and Density



The background of the slide features a photograph of modern architecture. On the right side, there is a tall building with a facade made of numerous vertical, light-colored slats or louvers. On the left side, there is a lower building with a similar slatted facade and visible windows. The sky is a clear, pale blue.

City Rail Link brings travel time savings for most Aucklanders when using public transport.

By making more places more accessible to more people, CRL can enable new developments, provide access to more jobs and enable Auckland to function more effectively as a well-connected city.

Public transport infrastructure can increase land values, but this does not necessarily lead to development, particularly in the short term.

This section looks at population density, building consents, changes in citywide access and the relationship between deprivation and access to jobs. It aims understand how CRL may influence development.

What should Auckland do?

Acknowledge that CRL will change demand patterns. Prioritise masterplanning in areas undergoing change. Monitor deprivation levels in areas with high public transport accessibility and use this information to understand how one may affect the other.

3. Development and Density

Population per km²



What are we measuring?

Population density across Auckland, measured by census area unit.

Why is it important?

Auckland has a long-standing ambition to flourish as a quality compact city. This means enabling higher residential population densities in existing urban areas.

Higher-density development requires higher-capacity public transport.

As part of an enhanced public transport network, CRL has the potential to enable higher residential densities, plus complementary land uses.

*For an excellent perspective on high-density development, see: *Where is the apartment demand in Auckland?* (CBRE, Sept 2025)

What should we expect?

Density is more likely to increase in rail catchments than in other areas, particularly in those neighbourhoods which already attract high-density residential development*.

This measure may take some years to be fully realised. This is because of the time that it takes to deliver high-density residential development.

What does the map show?

Population densities in Auckland are highest in the city centre and city centre fringe, where public transport connectivity is highest.

The average population density in Auckland is 4,335 residents per km². This holds for most areas of urban non-central Auckland, except for industrial areas.

Population density in Auckland in SA2 units

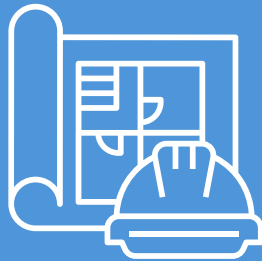


3. Development and Density

Building consents per km²

41%

The average proportion of dwelling consents in rail catchments over the last 10 years



4,394

The mean number of consents in rail catchments per year over the last 10 years

What are we measuring?

The number of building consents is a lead indicator of urban development activity in the rail catchments.

Building consents strongly correlate with completion and delivery of residential dwellings.

Research* shows that 90% of buildings consented are constructed, and most begin within six months.

Why is it important?

Due to the expected lag between dwellings being completed and CRL opening, consents provide us with an advanced indication of how the market is being affected.

What should we expect?

There may already be effects due to the announcement of CRL and its imminent completion.

This data will help to understand how businesses and people make decisions before and after major transport projects.

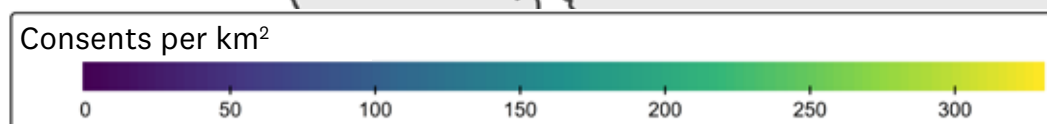
There are many other factors that affect consents both directly and indirectly. These include regulatory factors such as land use zoning and economic factors such as interest rates.

What does the map show?

The map shows the number of consents per square kilometre for each SA2 unit in Auckland.

See: OurAuckland (2025): Auckland's housing pipeline shows how consents become new homes

Building consents per km² in Auckland in SA2 units



3. Development and Density

Change in public transport access from anywhere to anywhere

10.2%

Reduction in total anywhere-to-anywhere travel time at Glen Eden



37%

Reduction in total anywhere-to-anywhere travel time at Karaka Flats

What are we measuring?

This map shows the change in anywhere-to-anywhere travel times by public transport across all of Auckland, pre and post-CRL.

Why is it important?

The time-saving effects of CRL are not evenly distributed. Places that are *already* easy to reach from much of Auckland (see p. 37) will see small percentage reductions in travel time.

On the other hand, previously-disconnected areas that receive brand new public transport infrastructure and services will see significant travel time reductions.

What should we expect?

Areas experiencing travel time reductions may attract significant development. There may already be effects from the announcement of CRL.

What does the map show?

Reductions in anywhere-to-anywhere travel times are greatest to the south and west.

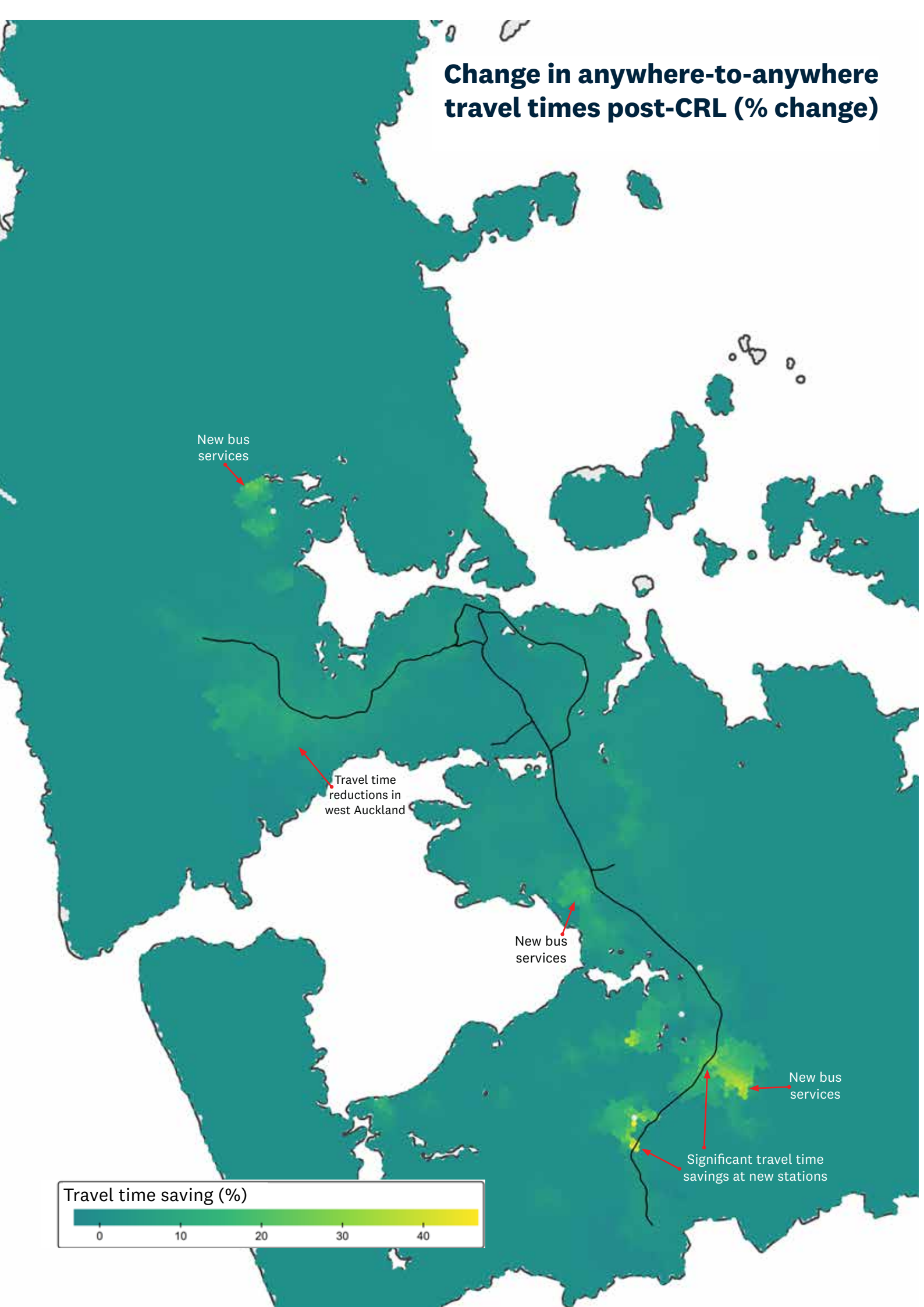
- **Western Line stations** see 5% -10%, reductions in anywhere-to-anywhere times, reflecting the more direct services of CRL.
- **Drury and Paerata** have the highest percentage reductions in anywhere-to-anywhere journey times (32% and 42% respectively) due to new stations.

A note on cumulative numbers

This map shows the total change in time taken to reach each hex area *from every other hex area in Auckland* (served by public transport).

The total CRL day-1 reduction in Auckland-wide travel time to Drury, for example is **83,770 hours**.

Change in anywhere-to-anywhere travel times post-CRL (% change)



New bus services

Travel time reductions in west Auckland

New bus services

New bus services

Significant travel time savings at new stations

Travel time saving (%)

0

10

20

30

40

3. Development and Density

Deprivation and access

5.2

The mean deprivation index for Auckland



6.6

The mean deprivation index for rail catchment (higher = more deprived)



What are we measuring?

Deprivation levels across Auckland (NZDep) compared with access to jobs by public transport (see p. 31).

Why is it important?

Auckland has significant levels of sub-urban poverty, typically associated with low levels of access to employment. CRL has the potential to help address poverty and deprivation by increasing access to jobs by public transport.

Increases in accessibility may change demand patterns, land values and development opportunities.

What should we expect?

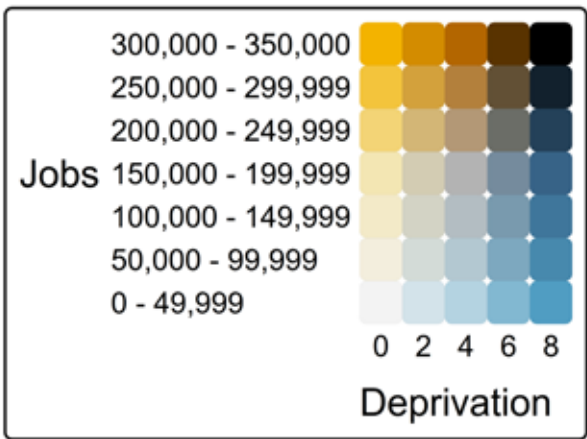
All things being equal, an increase in public transport accessibility will increase people’s access to jobs and reduce levels of deprivation

What does the map show?

It shows which parts of Auckland hav

- Low access; low deprivation
- Low access; high deprivation
- High access; high deprivation
- High access; low deprivation

Long-term monitoring may reflect changes in the distribution of these patterns over time.



**Bivariate plot of:
Deprivation levels (NZ Dep)
Access to jobs by public transport**

